AI-Assisted Customer Inquiry and Product Discovery in a Web-Based Bicycle Sales System: A Digital Innovation Proposal for FlowBikes

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Abstract

FlowBikes is a bicycle sales business for which the project proposes a web-based system to centralize bicycle listings and support customer product discovery. The organizational problem addressed is the fragmentation of product information and the effort required to respond to repeated questions about bicycle specifications and suitability. This paper proposes a public product catalog with administrator-managed product records and an AI Concierge that provides initial answers and product suggestions using approved product information. The study is proposal-stage: no completed user study, benchmark, sales analysis, or measured operational impact is claimed. A document-based analysis of project requirements and relevant literature informs the conceptual framework, feasibility assessment, risk controls, and a future evaluation protocol. The proposed evaluation covers answer relevance and factuality, usability, safe escalation, product-data quality, and operational feasibility. Responsible implementation would require accurate and maintained catalog data, privacy and cybersecurity controls, transparent AI limitations, human oversight, and an explicit route to seller assistance. The proposal contributes a context-specific evaluation plan for assessing whether AI-assisted discovery is useful and feasible for a small bicycle retailer.

Keywords: digital innovation; artificial intelligence; customer inquiry; product discovery; bicycle retail; responsible AI; feasibility

1. Introduction and Organizational Problem

Digital platforms can help small retail businesses organize product information and give customers a more convenient way to explore products. FlowBikes is a bicycle sales business offering bicycles and related products. The project context describes a need to centralize product listings and support customers who ask staff for specifications or guidance before deciding which bicycle may suit their needs. Repeated questions and fragmented product information can make it harder to keep product details consistent and help customers compare options efficiently. The extent of these effects at FlowBikes has not yet been measured and should be validated with business records or stakeholder feedback.

This paper proposes a web-based bicycle sales information system with a public catalog and an AI Concierge for initial customer inquiry support. The AI feature is intended to help customers navigate available product information, not to replace staff judgment or make final sales decisions. The project scope specifies administrator authentication only and excludes customer accounts and online payment processing. Customers may contact the seller directly for purchase discussions or confirmation.

1.1 Research Objectives

• Explain how an AI-assisted web catalog may address product discovery and customer inquiry needs at FlowBikes.

• Synthesize literature and guidance relevant to software risk management, project decision-making, and responsible AI governance.

• Present a conceptual framework linking the organizational problem, proposed technology functions, and expected outcomes.

• Assess technical, operational, financial, organizational, security, privacy, and ethical feasibility.

• Define an evaluation protocol and indicators that can be applied after implementation.

2. Related Literature and Research Gap

Software risk-management research emphasizes identifying and reviewing risks throughout the software life cycle rather than treating risk assessment as a one-time activity. Masso et al. (2020) review software risk-management practices across the life cycle, supporting staged review of integration, data quality, and AI response accuracy. Adriani et al. (2024) also examine risk-management practices across software development and discuss challenges related to resources, communication, and process complexity. For FlowBikes, these themes support a staged approach in which technical and organizational risks are reviewed during design, integration, testing, and any later pilot.

Decision-making in software project management also depends on communication, stakeholder participation, and traceable reasoning. Cunha et al. (2016) review decision-making themes in software project management. Applied to FlowBikes, the business owner and staff should help define useful product information, review the AI feature's boundaries, and decide whether proposed capabilities meet business needs. Decisions and changes should be documented rather than treating generated recommendations as automatically correct.

Micán et al. (2021) discuss project risk management and risk interdependencies in a broader project-management setting. This is relevant because product-data quality, AI response behavior, third-party service availability, testing time, and staff availability can affect one another. A failure in one component may reduce the reliability or usefulness of the wider workflow.

Responsible AI guidance is important because conversational systems can produce incomplete, inaccurate, or misleading answers. The NIST AI Risk Management Framework (NIST, 2023) organizes risk work around Govern, Map, Measure, and Manage. UNESCO's Recommendation on the Ethics of Artificial Intelligence (UNESCO, 2021) emphasizes human rights, transparency, accountability, and ethical governance. For FlowBikes, these principles support grounding answers in approved product information, testing difficult questions, minimizing personal-data collection, explaining AI limitations, and providing a clear route to a human seller.

2.1 Literature Synthesis and Research Gap

The literature converges on three themes: risk management should continue across the life cycle; project decisions benefit from stakeholder involvement and documented reasoning; and AI-related risks require explicit governance and evaluation. These sources provide general principles, but they do not establish the effectiveness of an AI Concierge for FlowBikes. The project-specific research gap is the absence of verified local evidence showing whether AI-assisted product discovery improves inquiry handling, information access, or staff workload in this business.

The appropriate next step is to establish a baseline and conduct structured scenario testing and, if feasible, usability evaluation with the business owner, staff, or consenting users. Until such evidence is collected, benefits must be presented as expected outcomes rather than demonstrated results.

3. Proposed Technology and Innovation Strategy

3.1 Proposed Technology

The proposed innovation is a web-based bicycle sales information system with a public product catalog and an AI Concierge. The catalog is intended to display bicycle listings and specifications, support browsing and filtering, and use administrator-managed records to keep product information current. The AI Concierge is intended to answer common product questions and suggest relevant bicycles based on needs stated by the customer. Responses should rely on approved product information, communicate uncertainty, and direct customers to staff when price, stock, fit, compatibility, or purchase details need confirmation.

3.2 Suitability and Strategic Approach

• Prioritize the core catalog and accurate product data before expanding AI-assisted features.

• Use approved product records as the source for factual claims and avoid inventing specifications, prices, or stock status.

• Test typical, ambiguous, out-of-scope, and adversarial questions, including cases where the system should say it cannot verify an answer.

• Provide a clear human escalation route for uncertain, sensitive, or purchase-specific questions.

• Monitor third-party API availability, usage limits, response quality, privacy, and security.

• Approve scope changes only after considering their effects on schedule, resources, quality, and cost.

3.3 Expected Organizational Value

Potential value includes easier access to bicycle specifications, more consistent product presentation, initial assistance with common questions, and a clearer path for handling inquiries that require human judgment. These benefits are hypotheses to evaluate. The proposal does not claim that sales, service speed, customer satisfaction, or staff workload have already improved.

4. Research Methodology and Evaluation Protocol

This is an applied, proposal-stage case analysis of a specific business information-system use case. The current analysis draws on FlowBikes project requirements and planning artifacts and a review of relevant scholarly literature and official guidance. No completed user study, operational benchmark, system performance test, or measured business outcome was included in the materials available for this finalization.

4.1 Proposed Evaluation Procedure

Confirm the approved product catalog fields, current product records, AI feature scope, and human escalation rules with the business owner or designated staff.

Prepare a test set of representative questions covering specifications, comparisons, rider needs, price or stock confirmation, ambiguous requests, unavailable information, and out-of-scope questions.

Have a reviewer compare each response with approved product records and score relevance, factuality, clarity, and safe handling of uncertainty using a documented rubric.

Record whether the response should have escalated to staff, whether it did so, and whether any unsupported product claim appeared.

If a prototype and suitable participants are available, conduct a small usability review with informed participation and collect feedback without unnecessary personal information.

Report the number and nature of tests, the scoring method, limitations, and actual results. Do not claim improvement without comparable baseline and follow-up evidence.

4.2 Proposed Evaluation Dimensions

All measures are proposed. Targets should be agreed with the owner after a baseline is established; no results are reported here.

5. Conceptual Framework

The proposed framework links the organizational problem to technology functions and expected outcomes. Fragmented product information and repeated inquiries are the problem context. The proposed functions are a centralized catalog, administrator-managed product records, filtering, AI-assisted question answering, product suggestions, and human escalation. Expected outcomes to evaluate include more accessible product information, relevant and factual answers, usable next steps, and operational feasibility. These relationships are propositions, not validated causal findings.

6. Feasibility Analysis

7. Ethical and Professional Considerations

7.1 Responsible AI and Human Oversight

The AI Concierge may generate incorrect specifications, outdated stock or price information, or unsuitable suggestions. Answers should be grounded in approved product records; uncertain questions should be escalated; and the system should not imply that recommendations are guaranteed expert advice. Testing should include edge cases and a documented process for reviewing failures. The NIST AI RMF and UNESCO guidance provide useful voluntary frameworks for risk management and ethical oversight.

7.2 Privacy and Data Protection

Although browsing a public catalog may not require personal information, free-text questions can contain names, contact details, locations, preferences, or other personal data. FlowBikes should minimize collection, explain the purpose of processing, restrict access, set appropriate retention practices, and review the terms of any external AI provider. Relevant obligations under the Philippine Data Privacy Act of 2012 (Republic Act No. 10173) should be assessed against actual data practices; this paper does not assert that compliance has been independently audited.

7.3 Cybersecurity, Governance, and Sustainability

Risks include compromised administrator credentials, exposed API keys, unauthorized product changes, malicious inputs, prompt manipulation, insecure database configuration, and third-party outages. Recommended controls include strong administrator authentication, least-privilege access, protected secrets, server-side validation, secure configuration, updates, backups, and appropriate logs. The business owner should approve scope and acceptance criteria; assigned staff should maintain product information; and the technical maintainer should manage deployment, access, defects, and service monitoring. Hosting, AI/API use, and maintenance create recurring costs, so usage and ongoing value should be reviewed.

7.4 Academic Integrity and Evidence Quality

Claims about literature, legal requirements, system performance, and FlowBikes operations must be supported by sources or project evidence. In-text citations must match the reference list. The group should not invent consultation feedback, test results, sales outcomes, or performance improvements. AI assistance is disclosed in this paper, and the group remains responsible for verifying references, factual claims, and final wording.

8. Limitations

This paper is a proposal-stage analysis. The materials reviewed did not establish a completed user study, benchmark test, verified budget, measured customer satisfaction, measured inquiry-handling time, or sales impact. Actual mock-colloquium comments and completed peer-review feedback were not provided in the available Week 13 documents; the supplied revision plan lists provisional improvement areas rather than recorded comments. Accordingly, the paper addresses those improvement areas by clarifying the research gap, specifying a test protocol, and making feasibility and safeguards more concrete, but the group must still incorporate any actual feedback received.

9. Conclusion and Recommendations

The proposed web catalog and AI Concierge may help FlowBikes centralize product information and provide initial assistance with bicycle inquiries. The solution appears feasible in principle, but actual feasibility depends on reliable product data, technical integration, staff participation, affordable services, security controls, and testing. The current evidence does not demonstrate that the system has improved business performance.

• Prioritize the core catalog and product-data management before expanding AI functions.

• Use approved product data, communicate uncertainty, and provide human escalation for questions that need confirmation.

• Apply privacy and cybersecurity safeguards based on actual data flows and system configuration.

• Agree on a test set and scoring rubric for relevance, factuality, clarity, and safe escalation.

• Establish baseline measures before deployment and report results transparently, including failures and limitations.

• Update this paper and presentation with actual mock colloquium feedback and the group's documented response before final submission.

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AI-Use Disclosure

ChatGPT was used to assist with organizing and revising the paper, synthesizing literature themes, refining the conceptual framework, developing a proposed evaluation protocol, and identifying feasibility and responsible-innovation considerations. The group must verify all source details against original publications or official sources, confirm FlowBikes-specific statements against approved project materials, and review the final text. The group remains responsible for the accuracy, originality, citations, and submission of this work.

Dimension	Proposed measure	Evidence / timing
Answer relevance	Share of responses judged to address the question asked.	Reviewed scenario test set before acceptance and after material changes.
Factuality	Share of factual product claims supported by approved catalog records.	Reviewer checks against current product data.
Safe escalation	Share of questions requiring confirmation that are correctly referred to staff.	Test cases for price, stock, compatibility, and uncertainty.
Usability	Reviewer or participant assessment of clarity and next steps.	Usability review or pilot, if feasible and consented.
Product data completeness	Share of active listings meeting a defined specification checklist.	Baseline and periodic catalog audit.
Inventory record accuracy	Share of sampled records matching verified shop records.	During testing and after deployment, if records are available.
Routine inquiry handling time	Time needed to answer a defined set of routine questions.	Before and after comparison only if comparable records exist.

Problem context	Proposed functions	Expected outcomes to evaluate	Conditions / moderators
Fragmented product information; repeated technical questions; difficulty comparing options.	Central catalog; product filtering; approved product records; AI Concierge; human escalation.	Information accessibility; answer relevance and factuality; usability; feasibility.	Product-data quality; AI behavior; staff adoption; internet and service reliability; user trust.

Dimension	Initial assessment	Requirements and risks
Technical	Potentially feasible as a web application, subject to actual configuration and team capacity.	Confirm data model, integration compatibility, API credentials, service limits, deployment, backups, error handling, and integration tests.
Operational	May fit product-listing and inquiry work if staff maintain records and review escalated questions.	Assign ownership for product updates, define escalation and response procedures, and collect staff feedback.
Financial	Cannot be confirmed without a verified cost estimate.	Estimate hosting, database, AI/API, and maintenance costs; review usage pricing and monitor actual costs.
Organizational	Depends on business participation and clear team responsibilities.	Agree on scope and acceptance criteria, schedule reviews, record decisions, and manage change requests.
Security and privacy	Customer prompts may contain personal information; administrator functions and integrations create security risks.	Minimize data, restrict access, protect secrets, validate inputs, secure configuration, and define retention and access practices.
Ethical AI	AI may produce incorrect, incomplete, or overconfident advice.	Ground answers in approved data, test edge cases, disclose limitations, allow human escalation, and review failures.
Schedule and resources	Academic deadlines and team availability constrain integration and testing.	Prioritize approved scope, track dependencies, reserve test time, and assess changes before accepting them.